

Microgrid Panel Technical Report

1. System Overview

This report details the configuration and material requirements for a customized Microgrid Panel. The system handles **2 DG(s)**, **120.0 kW** Grid supply, and **100.0 kWp** Solar PV. Managed via a centralized Microgrid Controller (MGC).

2. System Specifications

Total Busbar Current Rating: 1078.18A

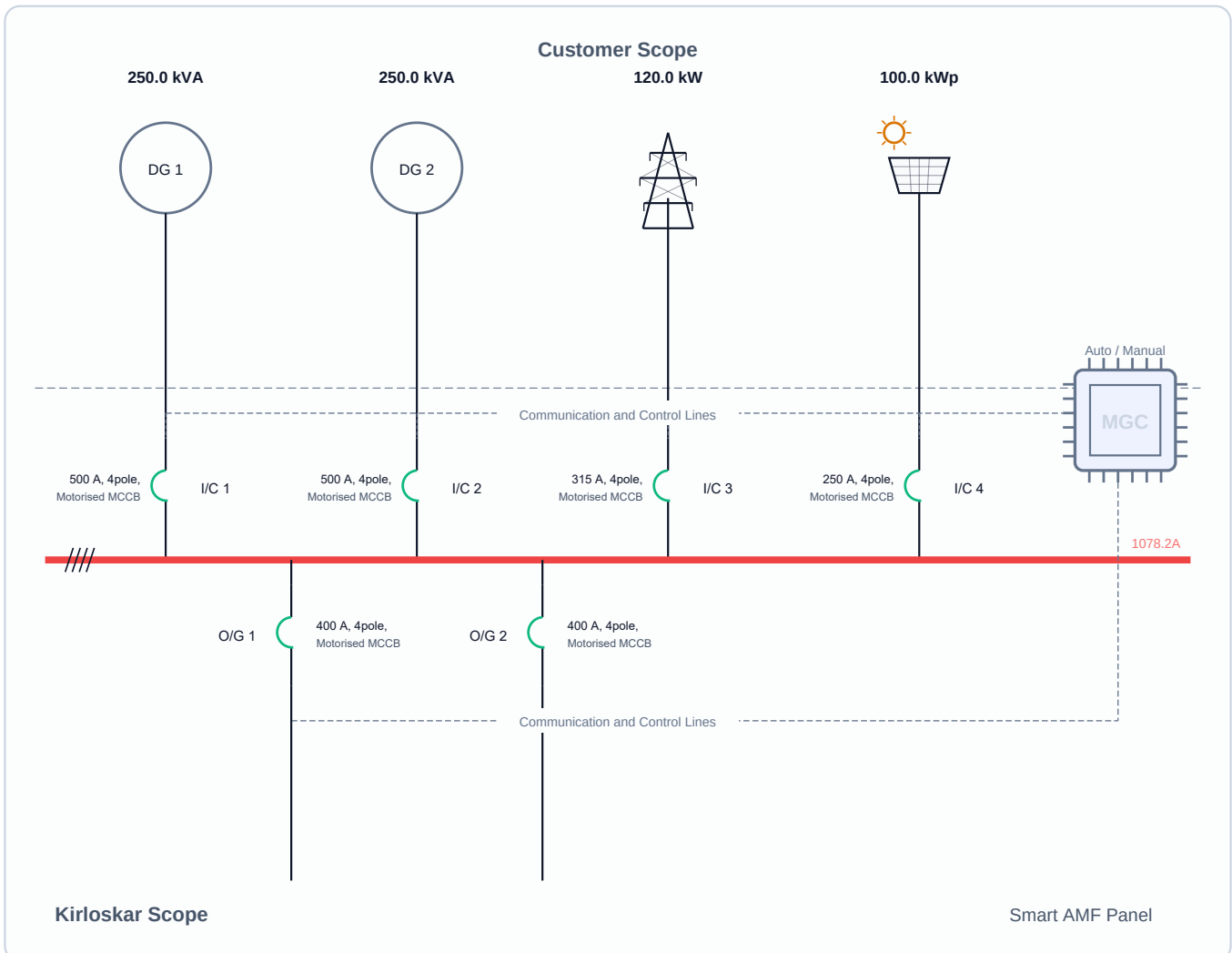
Total Outgoing Capacity: 800A

Recommended Busbar: 1 Set (110 x 20 mm Aluminium)

Panel Dimensions (Computed): 900W x 1500H x 400D mm

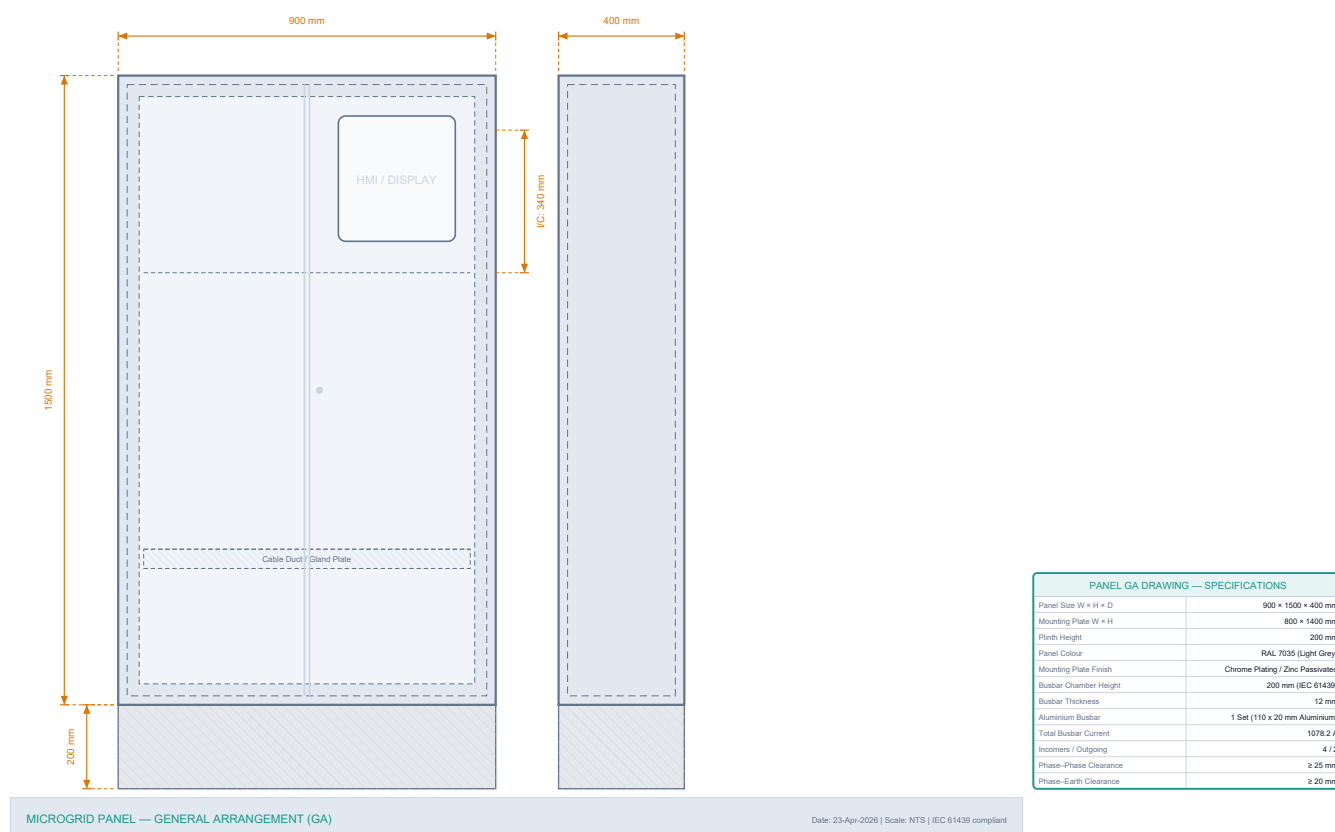
System Configuration: 4-Phase, 2 Outgoing Feeders

3. Single Line Diagram (SLD)



Note: Diagram illustrates electrical topology and power flow.

4. General Arrangement (GA) Drawing



Internal layout and dimensional overview.

4.1 MCCB Schedule (from Database)

Tag	Description	Rating (A)	Poles	H × W × D (mm)	Frame
I/C 1	Incomer MCCB	500A	4P	340×140×110	L-Frame
I/C 2	Incomer MCCB	500A	4P	340×140×110	L-Frame
I/C 3	Incomer MCCB	315A	4P	279×152×148	400A
I/C 4	Incomer MCCB	250A	4P	191×104×86	J-Frame
O/G 1	Outgoing MCCB	400A	4P	279×152×148	400A
O/G 2	Outgoing MCCB	400A	4P	279×152×148	400A

5. Bill Of Material (BOM)

Sr	Component / Description	Rating	Qty	UOM
1	MCCB 250A, 4P, 36kA	250A	1	Nos
2	MCCB 315A, 4P, 36kA	315A	1	Nos
3	MCCB 500A, 4P, 50kA	500A	2	Nos
4	MCCB 400A, 4P, 50kA	400A	2	Nos
5	Aluminium Main Busbar (110 x 20 mm Aluminium)	1078.2A	1	Set
6	Microgrid Controller (MGC)	Standard	1	Nos
7	Control Cable 1.5 sqmm	-	100	Meters
8	Power/Consumable Cable Varied	-	50	Meters
9	Control Panel, 1500H x 900 W x 400 D mm with stand	-	1	Set

6. Notes & Remarks

- This BOM is subject to final design review.
- All MCCB ratings include a 1.25× safety factor.
- Busbar sizing considers thermal conductivity and current density limits.
- Panel dimensions are computed dynamically from MCCB database.
- All components per relevant Indian Standards and IEC guidelines.